

Design Thinking & Industry 4.0

Modules 3 and 4, compressed to what an MCQ can actually ask. Cover each section, try to recall it, then uncover and check.

MODULE 3 — DESIGN THINKING

Definition. A **human-centric** approach to innovation that starts from what real users need, not from what the company can build. It is **iterative** and **non-linear**.

Names to know: popularised by **IDEO** and its CEO **Tim Brown** (*Change by Design*, 2019 — on your syllabus). The five-stage model comes from **Stanford’s d.school**.

The five stages — most likely question on the paper

Empathise → Define → Ideate → Prototype → Test

STAGE	WHAT HAPPENS	OUTPUT
1. Empathise	Understand users — observe, interview, immerse. No solutions yet.	User insights
2. Define	Frame the real problem	Problem statement / POV
3. Ideate	Generate many ideas — brainstorming, SCAMPER	A pile of ideas
4. Prototype	Build a quick, cheap, scaled-down version	A testable model
5. Test	Users try it, feedback gathered, then loop back	Refinements

TRUE / FALSE TRAP

“Design Thinking is a strictly linear process” → **FALSE**. You may return to any earlier stage at any time.

The three lenses

A good solution sits where all three overlap: **Desirability** (do people want it? — human) · **Feasibility** (can we build it? — technical) · **Viability** (can it sustain itself? — business).

Divergent vs Convergent thinking

Divergent — open up, generate *many* options (Ideate). **Convergent** — narrow down, *select* the best (Define, and after Ideate). Design Thinking alternates between the two.

Empathy mapping — four quadrants

SAYS · THINKS · DOES · FEELS

Says and **Does** are *observable*. **Thinks** and **Feels** must be **inferred** — you cannot see a thought. (*Frequently asked.*) Other Empathise-stage tools: user interviews, observation, personas, journey maps.

Problem framing (Define stage)

- **POV statement:** [User] needs [need] because [insight]
- **“How Might We...” (HMW)** — reframes a problem as an opportunity
- A good problem statement is human-centred, broad enough for creativity, narrow enough to act on

Brainstorming rules

Defer judgement · **Go for quantity** · Encourage wild ideas · Build on others’ ideas · Stay on topic · One conversation at a time · Be visual.

TRUE / FALSE TRAP

“Weak ideas should be criticised immediately during brainstorming” → **FALSE**. Judgement is deferred; quantity beats quality at this stage.

SCAMPER — all seven

LETTER	MEANS	EXAMPLE
S	Substitute	plastic → bamboo straw
C	Combine	phone + camera
A	Adapt	velcro, copied from burrs
M	Modify / Magnify / Minify	iPad → iPad Mini
P	Put to another use	baking soda as a cleaner
E	Eliminate	wireless earbuds — remove the cable
R	Reverse / Rearrange	pay-after-delivery

Prototyping & testing

- **Low-fidelity** = rough, cheap, fast (paper, cardboard). **High-fidelity** = close to the real product.
- Purpose: **fail fast, fail cheap** — learn before spending money
- **Iterative feedback loop:** build → test → refine → repeat

Benefits: user satisfaction, reduced risk, encourages innovation, cross-functional teamwork. **Limitations:** time-consuming, needs skilled facilitation, results hard to measure.

MODULE 4 — TECHNOLOGY, TRANSFORMATION & INDUSTRY 4.0

The four industrial revolutions

	ERA	DRIVEN BY
Industry 1.0	~1784	Steam power , water power, mechanisation
Industry 2.0	~1870	Electricity , mass production, the assembly line
Industry 3.0	~1969	Electronics, computers, IT , basic automation
Industry 4.0	2011 →	Cyber-physical systems, IoT, AI, cloud

The term “**Industry 4.0**” was coined in Germany in 2011 (Hannover Fair). **Industry 5.0** is the emerging next step — human-machine collaboration, sustainability, personalisation.

The nine pillars of Industry 4.0

Big Data & Analytics · Autonomous Robots · Simulation (Digital Twin) · System Integration · Industrial IoT · Cybersecurity · Cloud Computing · Additive Manufacturing (3D printing) · Augmented Reality

Core terms

TERM	DEFINITION
IoT	Internet of Things — network of physical devices with sensors , connected to the internet, exchanging data
IIoT	Industrial IoT — IoT applied inside factories
Digital Twin	A live virtual replica of a physical asset or process, updated in real time by sensors
Cyber-Physical System	Integration of computation with physical processes — the backbone of Industry 4.0
Big Data	Data too large/complex for traditional tools. 3 Vs: Volume, Velocity, Variety (5 Vs add Veracity, Value)
Cloud computing	On-demand delivery of computing resources over the internet
Smart factory	A fully connected, self-optimising production facility
RPA	Robotic Process Automation — software bots doing repetitive rule-based tasks
Additive manufacturing	3D printing — building by <i>adding</i> layers (subtractive = cutting away)
Blockchain	Decentralised, tamper-proof distributed ledger — supply-chain traceability

AR vs VR

AR = your PHONE. Camera up, you see your *actual room* with something added (Pokémon Go, Instagram filters). Reality is still there.

VR = a HEADSET over your eyes. Your room is *gone*, replaced by a virtual world.

Phone → AR. Headset → VR. If you can still see the real world, it is AR.

Cloud service models

MODEL	PROVIDER GIVES YOU	YOU ARE A...	EXAMPLE
IaaS	Hardware only — you install the OS and app	Sysadmin	AWS EC2
PaaS	Hardware + OS — you deploy your own code	Developer	Heroku, App Engine
SaaS	Everything, including the app	End user	Gmail, Netflix, Zoom

Deployment models: public · private · hybrid.

EASY TO GET WRONG

Gmail is SaaS, not PaaS. The “platform” in PaaS is a *development* platform for building your own software — not the everyday sense in which Gmail is “a platform”. SaaS also has nothing to do with subscriptions: Gmail is free and still SaaS.

Pizza version: IaaS = frozen pizza in **your** oven · PaaS = delivery to **your** table · SaaS = you go to a **restaurant**.

Digital transformation

Integrating digital technology into **all areas** of a business, fundamentally changing how it operates and delivers value. A **digital roadmap** is the phased plan for getting there.

TRUE / FALSE TRAP

“Digital transformation means simply purchasing new software” → **FALSE**. It requires culture, process and business-model change too.

Sustainable operations

- **Triple Bottom Line: People · Planet · Profit** (the 3 Ps)
- **Circular economy** — reduce, reuse, recycle; design out waste. Opposite of the linear “take-make-dispose” model
- **ESG** — Environmental, Social, Governance
- **SDGs** — UN Sustainable Development Goals (**17**)
- **Green computing** — energy-efficient IT

Ethical issues in digital services

Data privacy (GDPR) · algorithmic bias · job displacement through automation · transparency and accountability · cybersecurity and consent.

Sector applications

SECTOR	TYPICAL APPLICATION
Manufacturing	Predictive maintenance, smart factories
Retail	Personalised recommendations, inventory forecasting
Healthcare	Telemedicine, AI diagnostics, wearables
Hospitality	Chatbots, digital check-in, dynamic pricing
Public service	E-governance, digital ID, smart cities

THE FIVE YOU ACTUALLY GOT WRONG TONIGHT

- 1

Digital Twin = a live virtual replica of a *physical thing*, not a cloned database.
- 2

Gmail / Netflix / Zoom = SaaS, not PaaS. You are an end user, not a developer.
- 3

AR adds to reality (phone). VR replaces it (headset).
- 4

IoT = Internet of Things — physical objects, not “technology”.
- 5

Empathy map: Thinks and Feels are *inferred*; Says and Does are *observed*. (True, not false.)

Built during the session of 25–26 Sep 2026 · Learning Spot workspace.
Syllabus reference: Brown, T. (2019). *Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation*. Harper Business.
Anything unclear? Ask your teacher — that’s me. Just ask in the chat.